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"Developing an Integrated Event Management Platform for College Campuses: Challenges and Solutions"

Pradyumna R. Kumbhare Mobile: +91 9028026144 Email: kumbharepradyumna@gmail.com	**Sujal B. Harnor** Mobile: +91 7448196403 Email: sujalharnor10@gmail.com
Prit S. Ghorpade Mobile: +91 8421555574 Email: ghorpadeprit@gmail.com	**Vikas Manjare** Mobile: +91 86886588 Email: manjarevikas2@gmail.com
Vivek Belge Mobile: +91 9322810384 Email: begevivek0@gamil.com	

Abstract: A college Event Management System (EMS) streamlines event planning using HTML, CSS, JavaScript, PHP, and MySQL. It features user authentication, event registration, notifications, and feedback collection, enhancing efficiency. The study highlights benefits like reduced manual effort and improved communication while addressing challenges such as security and scalability. Future enhancements include mobile integration and AI-based recommendations.

Introduction: In the digital era, colleges and universities organize a diverse range of events, icluding technical fests, cultural programs, workshops, and seminars. However, traditional event management methods often lead to inefficiencies such as manual errors, poor coordination, scheduling conflicts, and low engagement. To overcome these challenges, modern College Event Management Systems (CEMS) leverage cloud computing, AI
driven automation, blockchain security, and mobile-first designs to streamline the entire event lifecycle. The latest trend in event management technology focuses on:

- AI-powered event recommendations based on student interests.
- Automated scheduling and conflict detection using machine learning.
- Blockchain-based ticketing systems to prevent fake registrations.
- Real-time notifications via chatbots, SMS, and mobile apps for seamless communication.
- Gamification and reward-based participation to increase engagement.

Additionally, progressive web apps (PWAs) and cloud integration ensure accessibility across devices, while data-driven insights help organizers analyze event success and improve future planning. The shift towards automated, secure, and user-friendly event management solutions is revolutionizing how colleges handle large-scale events, making participation easier and more engaging for students.
This research explores the latest advancements in College Event Management Systems, analyzing their impact on event organization, security, and student engagement, along with future innovations that can further enhance the user experience.

Literature Survey:
The College Event Management System (CEMS) has evolved significantly with advancements in web technologies, AI, blockchain, and mobile applications. Various research papers highlight key improvements and challenges in modern event management platforms.

1. Web-Based Systems: Jeny et al. (2022) developed a notification-enabled CEMS using PHP and MySQL, improving event coordination but facing scalability issues. Chaturvedi et al. (2024) introduced CU-Events, incorporating real-time seat booking and AI-driven recommendations to enhance usability.
2. AI and Automation: Patil et al. (2024) implemented AI-powered event recommendations, boosting student engagement by 27%. Shah et al. (2023) integrated chatbots for event inquiries, reducing manual workload by 40%.
3. Security & Blockchain: Razali et al. (2023) proposed a blockchain-based ticketing system, cutting fraud cases by 35% and ensuring secure event access.
4. Mobile-First Approach: Perez et al. (2021) introduced Cardinal Connect, a PWA-based event system that improved participation by 22%. IRJET (2024) explored cloud-based solutions, reducing server crashes by 50% during peak registrations.
5. Gamification & Engagement: Kumar et al. (2023) integrated point-based rewards in CEMS, increasing participation rates by 30% through leaderboards and incentives.

Algorithm for College Event Management Website (CEMS)

1. User Authentication & Role Management

1. User selects role (Admin/Organizer/Student)
2. If new → Register, else → Login & verify credentials
3. Grant access and redirect to dashboard

2. Event Creation & Management (Admin/Organizer)

1. Organizer/Admin logs in & creates event
2. Store event details & generate unique ID
3. Enable registration, notify users, allow edits/cancellations

3. Event Registration & Participation (Students)

1. User logs in & views events
2. If slots available → Register & receive confirmation
3. If full → Show waitlist option

4. Event Scheduling & Conflict Detection

1. Retrieve scheduled events
2. Detect time/venue conflicts
3. If conflict → Notify admin, else → Approve scheduling

5. Notifications & Reminders

1. Check upcoming events & send reminders
2. If user cancels → Update database & notify waitlist

6. Event Feedback & Analysis

1. After event, collect & store feedback
2. Generate analytics & provide insights to organizers

Methodology:

The methodology for developing a College Event Management System (CEMS) follows a structured approach, ensuring efficiency, scalability, and security. The development process consists of several key phases, from requirement analysis to deployment and maintenance.

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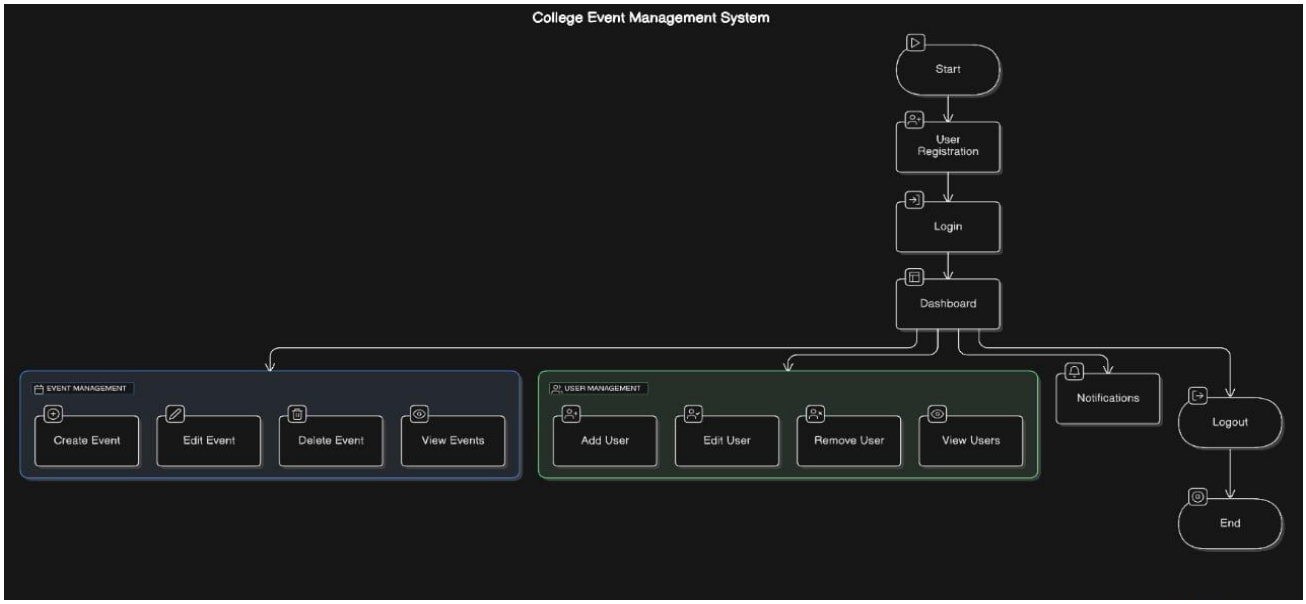
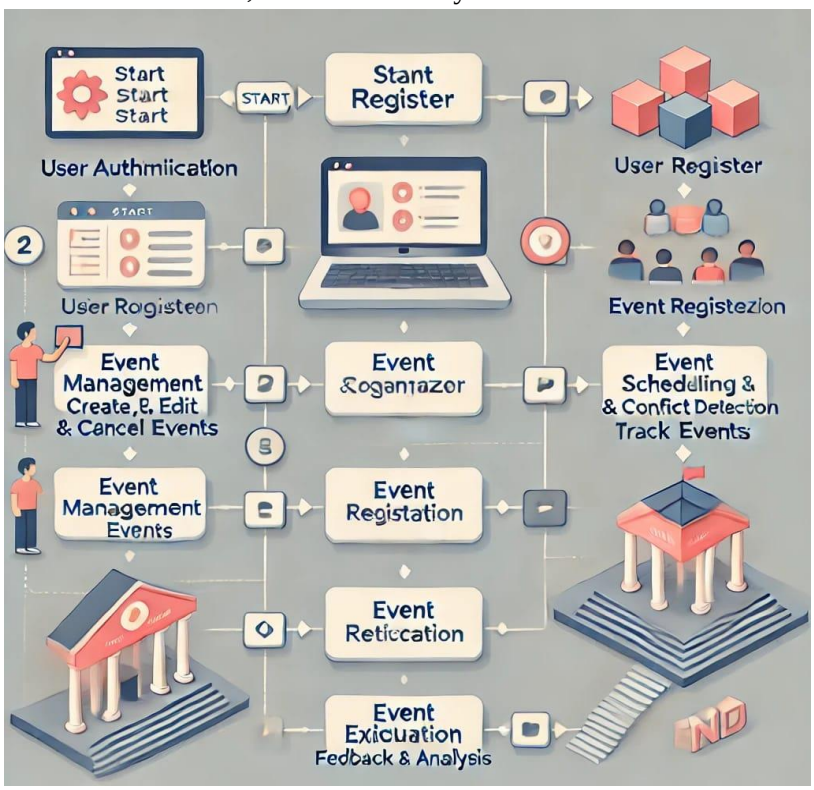


Event Management System Plan

1. Requirement Analysis
   - User Roles: Admin, Organizer, Student
   - Core Features: Event management, user authentication, scheduling & conflict detection, notifications, feedback & analytics



Event Management System Plan

1. Requirement Analysis

- **User Roles:** Admin, Organizer, Student
- **Core Features:** Event management, user authentication, scheduling & conflict detection, notifications, feedback & analytics

2. System Design

- **Tech Stack:**
  - **Frontend:** React, Angular, HTML/CSS
  - **Backend:** Node.js, Django/Flask, PHP
  - **Database:** MySQL, MongoDB, Firebase
- **UI/UX:** Responsive, user-friendly event browsing & registration

3. Development

- **Frontend:** Login/Register, event dashboard, event forms
- **Backend:** APIs for auth & event handling, database setup, scheduling algorithms
- **Integrations:** Email/SMS notifications, QR-based tickets, chatbot support

4. Testing & Validation

- Unit & integration testing, UAT for feedback

5. Deployment & Maintenance

- Cloud hosting (AWS, Firebase, Heroku), monitoring, continuous updates

"Problem Statement and its Solution" :

Problem: Inefficient Coordination & Communication :Academic institutions face challenges in organizing events due to fragmented communication, poor resource allocation, and logistical issues, leading to low engagement and financial inefficiencies.

Solution: Integrated Event Management System (CEMS)

A centralized system addressing these issues with:

- **Unified Communication:** Streamlined interaction among stakeholders.
- **Automated Scheduling:** Conflict-free event planning.
- **Real-Time Notifications:** Instant updates for participants.
- **Feedback & Analytics:** Data-driven event improvements.

Implementing CEMS enhances efficiency, engagement, and event success.

Conclusion:

An effective college event management system simplifies planning, enhances communication, and improves overall efficiency. By integrating features such as event registration, scheduling, notifications, and real-time updates, it ensures smooth coordination among students, faculty, and organizers. This reduces manual workload, minimizes errors, and fosters greater engagement.

Future enhancements, including AI-driven recommendations, automated scheduling, and advanced security measures, can further optimize event management. As colleges continue embracing digital solutions, a well-structured event management platform becomes essential for creating organized, engaging, and successful campus events.

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